

JAZZ VOICE CORES

VOIP VIDEOPHONE PROCESSORS



Video
Phones



VoIP in
Digital Displays



Portable
Handsets



IP
Webcam

**Highly efficient cores enabling
rapid integration of complete
speech processing.**

The Jazz VOIP Videophone Processor (VID) cores are highly optimized solutions specifically tailored to the demanding requirements of real-time video phones. Applications for supporting 2- or 3-way videoconferencing in desktop phones or integrated devices are moving into the market. The Jazz Voice VID cores can get you to market quickly and efficiently. They support a wide range of video and speech standards and are optimized for the high quality processing at a low cost. The VID cores are part of the broad Jazz Voice family from Improv.

Once the domain of high-cost equipment, low cost videoconferencing is becoming a reality. New standards like MPEG-4, H.264 and Windows Media are enabling consumer-friendly use of video telephony. The Jazz Voice VID solutions support a complete set of video and speech software modules that can be loaded from external memory; this allows a single core to support a broad range of standards.

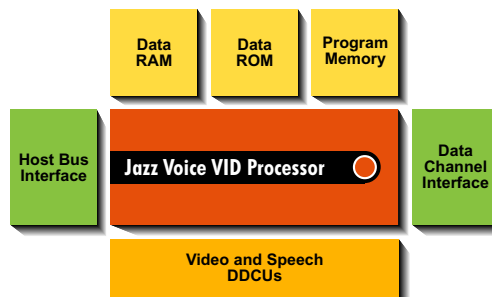
The Jazz Voice VID cores provide a complete package of integrated video and speech processing. The cores are pre-configured and pre-optimized and come with a complete suite of software modules. The cores have built-in interfaces for standard on-chip buses and memory management to allow direct drop-in into an existing SoC architecture.

OPTIMIZED CORES

The Jazz VID cores can be used as a black box implementation of a complete video and voice processing subsystem on a chip. Each core can operate as a standalone block or as a slave to a host processor.

Each VID core has a single, application-specific processor that combines a general-purpose instruction set with application-specific instructions to accelerate video and speech processing. Each core contains a very-long instruction word (VLIW) processor built using Improv's Jazz Composer tools.

In addition to the processor, the hardware core contains embedded memory, a host bus interface (HBI) that can connect to most standard on-chip buses including AMBA, OCP and Wishbone.



KEY FEATURES

- Complete silicon-ready solution
- Optimized for multiple standards and multiple channels
- Complete telephony framework for a drop-in silicon solution
- Available in multiple configurations
- Fully synthesizable and foundry independent
- Supported by a comprehensive software development tool suite
- Optionally available as configurable cores for additional customization

VID SOFTWARE MODULES

The VID software modules have been designed to fully exploit the unique advantages of the VID architecture.

- MPEG4 Video Codec
- H.264 Video Codec
- G.711 Speech Codec
- G.726 Speech Codec
- G.72x Speech Codecs
- Speech and Video Synchronization
- Comfort Noise Generation and Voice Activity Detection
- Caller-Id Processing
- Dial-Tone Management and Call Progress Tones
- Conferencing
- V.18 Hearing Impaired Support

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AVAILABLE PRODUCTS

The Jazz Voice VID family consists of the following specific products:

- **VID Webcam Core** is optimized for 2 channels of speech codecs and encoding of a single video.
- **VID Desktop Core** is optimized for 2 channels of speech and video codecs.
- **VID 4-Way Core** is optimized for up to 4 channels of speech and video codecs with integrated 4-way display.

Each core comes with complete software framework to support full video and speech synchronization.

IMPLEMENTATION CHARACTERISTICS

	<u>GATE COUNT</u>	<u>CLOCK</u>
VID Webcam	xxxK gates	40 MHZ
VID Desktop	xxxK gates	60 MHz
VID 4-Way	xxxK gates	120 MHz

These numbers represent the maximum requirements for running the VID software modules in a TSMC 0.13um process. The Jazz Voice VID cores have been highly optimized for high performance at low clock rates and low power.

- Fully synchronous design
- Silicon-proven in 0.18um and 0.13um technologies
- Advanced power management features
- Hardware Debug Environment through JTAG port
- Support for prioritized interrupts
- Full scan with BIST for test
- Boot-loadable firmware and smart caching (Jazz Jukebox)
- No custom physical components required

DEVELOPMENT TOOLS

A complete software development environment called the Jazz Standard tool suite supports Jazz Voice cores. This tool suite includes compiler, debugger, assembler, cycle-accurate instruction-set simulator and profiler/analyzer. The environment runs on Windows XP Professional, Windows 2000 and Windows NT environments.

The cores also come with a complete implementation environment called the Jazz Generator which creates complete synthesizable RTL, EDA tool scripts, verification environment and FPGA mapping.

As a separately purchasable item, Improv offers a development platform using Improv's FPGA-Based Jazz Rehearsal cards with a full verification environment to enable verification and system integration.

DELIVERABLES

- VID Processor Core
- VID Telephony Software Modules
- Verification Environment
- Jazz Standard Tool Suite
- Jazz PSA Generator
- User's Guide
- Implementation Guide
- Programmer's Guide
- Jazz Composer Tool Suite (for configurable package only)

LICENSING INFORMATION

The Jazz Voice VID cores are available today and can be licensed from Improv Systems. For additional information regarding the Jazz Voice family and other licensable Improv technology, please contact Improv Systems at sales@improvsys.com, or visit our website at <http://www.improvsys.com>. A list of worldwide local representatives can be found on the website.

ABOUT IMPROV SYSTEMS, INC.

Improv Systems, Inc. develops and licenses pre-configured and configurable Jazz Media, Jazz Voice and Jazz DSP cores. Improv's Jazz processor is a configurable DSP that allows designers to create optimized DSP cores targeted to consumer electronics and telecommunications markets. For more information, please visit www.improvsys.com.

Improv Systems, Inc. Corporate Headquarters
100 Cummings Center, Suite 341C
Beverly, MA 01915

phone: (978) 927-0555 • fax: (978) 927-0999
www.improvsys.com • sales@improvsys.com

WWW.IMPROVSYS.COM • SALES@IMPROVSYS.COM • PHONE: 978-927-0555

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